#HW 5

Graded

Student

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Total Points

100 / 100 pts

Question 1

1 **10** / 10 pts

✔ **– 0 pts** Correct

– 2 pts You should find out the compressibility factor in both cases

– 2 pts Compressibility factor calculation is not correct.

Question 2

2 **10** / 10 pts

✔ **– 0 pts** Correct

– 2 pts The temperature has been calculated incorrectly.

– 2 pts The work calculation is wrong.

– 3 pts incomplete

Question 3

3 **15** / 15 pts

✔ **– 0 pts** Correct

– 2 pts Heat transfer had to be calculated in kJ.

– 2 pts Work calculation is wrong.

– 2 pts The heat transfer calculation is wrong. Look into your del(U) calculation.

– 3 pts incomplete

Question 4

4 **15** / 15 pts

✔ **– 0 pts** Correct

– 2 pts W is correct, but Q is wrong. Del(U) will not be 0 here.

– 4 pts The process is right, but the calculation is wrong.

– 2 pts The heat transfer calculation is wrong.

Question 5

5 **15** / 15 pts

✔ **– 0 pts** Correct

– 2 pts W12 at constant pressure has to be calculated through PdV.

– 2 pts You have not calculated Q31 and W31

– 4 pts Your calculations are wrong.

– 6 pts incomplete

– 10 pts Late

Question 6

6 **35** / 35 pts

✔ **− 0 pts** Correct

Question assigned to the following page: 1

Homework 5

Monday, October 6, 2025 9 09 AM

$\boxed{3.21.67}$

a) Water $T_c = 647.3\,K$ $P_c = 220.9$ bar

$(600°F - 32) \times \frac{5}{9} + 273.15 = 588.706\,K$

$T_R = \frac{T}{T_c} = \frac{588.7}{647.3} \approx 0.91$

@ 900 lbf/in² = 62.05 bar

$P_R = \frac{62.05}{220.9} = 0.28$

$Z \approx 0.42$

Z not close to 1 not applicable

@ 100 lbf/in² = 6.89 bar

$P_R = \frac{P}{P_C} = \frac{6.89}{220.9} = 0.031$

$Z \approx 0.98$

Z close to 1 ideal gas model applicable

b) Nitrogen $T_c = 126\,K$ $P_c = 33.9$ bar

$-20°C + 273.15 = 253.15\,K$

$T_R = \frac{T}{T_c} = \frac{253.15}{126} \approx 2.0$

@ 75 bar

$P_R = \frac{75\text{ bar}}{33.9\text{ bar}}$

$= 2.21$

$Z \approx 0.97$

@ 1 bar

$P_R = \frac{1\text{ bar}}{33.9}$

$P_R = 0.029$

$Z \approx 0.99$

Both Z values close to 1 both scenarios applicable to ideal gas model

3.21.74

State 1: $p_1 = 30\ lbf/in^2$ State 2: $p_2 = 20\ lbf/in^2$

$T_1 = 540°F = 1000° R = 555\ K$ $V_2 = 4.5\ ft^3$

$V_1 = 4.0\ ft^3$

(a) $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ $T_2 = T_1 \frac{P_2 V_2}{P_1 V_1} = 1000 \times \frac{20 \times 4.5}{30 \times 4.0}$

$\boxed{T_2 = 750°R} = 416.67$

(b) $\Delta U = Q - W$ $m = \frac{p_1 V_1}{RT_1} = \frac{(4320)(4.0)}{53.35 \times 1000}$

$W_{total} = W - 1$ $\Delta U = -12 - (W-1)$ $m = 0.3238\ lbm$

$Q = -12\ Btu$ $\boxed{\Delta U = -11 - W}$

$-14.2 = -11 - W$ $\Delta U = m(298 - 400) = -14.98\ kJ$

$\boxed{W_{ork} = 3.2\ Btu}$ $= -14.2\ Btu$

3.21.76

State 1: $p_1 = 2\ bar = 2\ kPa$ State 2: $p_2 = 8\ bar = 8\ kPa$

$V_1 = 200\ K$ $V_2 = 2L = 0.002\ m^3$

$T_1 = 1L = 0.001\ m^3$ $R = \frac{8.314}{28.97} = 0.287$

$m = \frac{p_1 V_1}{RT_1} = \frac{200(0.001)}{(0.287)(200)} = 0.0035\ kg$

linear p-V → $p = m(V) + b$ → $800 = (0.002)m + b$

$200 = (0.001)m + b$

$p(V) = 600{,}000V - 400$ $600 = m(0.001)$ $\boxed{m = 600000\ kPa/m^3}$

$200 - 600{,}000 \times 0.001 = b = -400\ kPa$

$W = \int_{V_1}^{V_2} p\,dV = \int_{0.001}^{0.002} (600{,}000V - 400)\,dV$ $T_2 = \frac{P_2 V_2}{mR} = \frac{800(0.002)}{(0.0035)(0.287)} = 1600\ K$

$W = 300000V^2 - 400V \Big|_{0.001}^{0.002}$ $\Delta U = (0.0035)(1298.3 - 142.56) = 4.045\ kJ$

$Q = \Delta U + W = 4.045 + 0.5$

$\boxed{W = 0.5\ kJ}$ $\boxed{Q = 4.55\ kJ}$

Question assigned to the following page: 4

$\boxed{3.21.93}$ $W = \int_{0.1}^{0.3} P\,dV$

$m = \frac{P_1 V_1}{R T_1}$

$W = 2(0.3 - 0.1) = 0.4 \times 100 = 40\ kJ$

$m = \frac{200(0.1)}{(0.287)(300)}$

$W = 40/0.23 = 172.2\ kJ/kg$

$\boxed{W_{1-2} = 172.2\ kJ/kg}$

$m = 0.23\ kg$

$p = \frac{mRT_2}{V}$

$W_{2-3} = \int_{V_2}^{V_3} \frac{mRT_2}{V}\,dV$

$T_2 = T_1 \frac{V_2}{V_1} = 300\left(\frac{0.3}{0.1}\right)$

$= mRT_2 \ln \frac{V_3}{V_2}$

$T_2 = 900\ K$

$W_{23} = (0.23)(0.287)(900) \times \ln(2)$

$W_{23} = 41.58/0.23 = 178.9\ kJ/kg$

$\Delta U = 3(674.58 - 214.07)$

$\boxed{W_{23} = 178.9\ kJ/kg}$

$\Delta U = 460.5\ kJ/kg$

$Q = \Delta U + W_{total}$

$Q = 460.5 + 351$

$\boxed{Q = 811.5\ kJ/kg}$ $\boxed{W_{total} = 351\ kJ/kg}$

Question assigned to the following page: 5

$\boxed{3.21.95}$

$$R = \frac{(1.99 \times 778.17)}{28.97} \approx 53.35$$

$$v_1 = \frac{RT_1}{P} = \frac{53.35(500)}{(20)(144)} = 9.262 \text{ ft}^3/\text{lb}$$

$$Pv = RT \rightarrow \frac{T_2}{T_1} = \frac{v_2}{v_1} = 1.4$$

$$T_2 = 1.4 \times 500 = 700°R$$

$$P_3 = \frac{RT_3}{v_3} = \frac{53.35(920)}{9.262} \Big/ 144$$

$$P_3 = 32.8 \text{ psia}$$

$$W_{12} = \int p\,dv = p(v_2 - v_1)$$

$$= 2880(12.97 - 9.262)$$

$$= 3.705$$

$$\boxed{W_{12} \approx 13.71 \text{ Btu}}$$

$u_1 \approx 198$ kJ/kg $= 85.24$ Btu/lb

$u_2 \approx 278$ kJ/kg $= 119.5$ Btu/lb

$u_3 \approx 327$ kJ/kg $= 140.43$ Btu/lb

Process 1-2

$$\Delta U_{12} = 119.5 - 85.2 = 34.3 \text{ Btu}$$

$$Q = 34.26 + 13.71$$

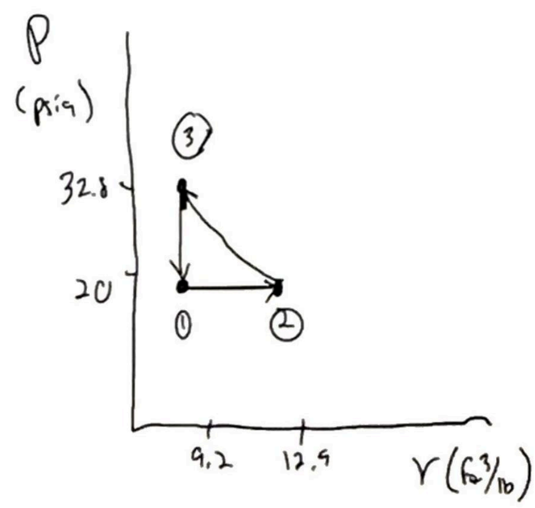
$$\boxed{Q_{12} = 47.92 \text{ Btu}}$$

Process 2-3

$$\Delta U_{23} = 140.43 - 119.5 = 20 \text{ Btu}$$

$$\boxed{Q_{23} = 0}$$

$$W_{23} = -\Delta U_{23} = \boxed{-20.93 \text{ Btu} = W_{23}}$$



$$\Delta U_{31} = 85.2 - 140.4 = -55.19 \text{ Btu}$$

Process 31

$$\boxed{W_{31} = 0}$$

$$\Delta U = Q \qquad \boxed{Q = -55.19 \text{ Btu}}$$